Addis Ababa Science and Technology University

Department of Mathematics, Physics and Statistics

Mathematics for Natural Sciences (Math 1007)

Group Assignment (10%)

Date: November 19, 2025

Instructions

Solve each of the problems in groups of up to 5 and submit your work by November 25, 2025. You will then be required to present randomly, and random questions will be forwarded to each member of the group. Your score will depend on your individual performance.

1. Name _____ Id.No _____ Signature _____
2. Name _____ Id.No _____ Signature _____
3. Name _____ Id.No _____ Signature _____
4. Name _____ Id.No _____ Signature _____
5. Name _____ Id.No _____ Signature _____

1. Let p, q, and r be propositions.

   (a) Use a truth table to determine whether the following compound proposition is a tautology, a contradiction, or a contingency:

   $$(p \Rightarrow q) \Rightarrow (\neg r \Rightarrow \neg p)$$

   (b) Using the laws of logic (such as De Morgan's laws, implications, associativity, etc.), prove that the following statement is a tautology *without* using a truth table:

   $$[\neg p \wedge (p \vee q)] \Rightarrow q$$

   (c) Negate the following statement and simplify it so that the negation symbol(s) only appear before the atomic propositions p, q, and r:

   $$(p \wedge q) \Rightarrow (r \vee \neg q)$$

2. Compute and discuss the truth value of the following:

   (a) $(\exists x)(\exists y)(x + y \neq y + x)$

   (b) $(\exists x)(\forall y)(x + y = 20)$

   (c) $(\forall y)(\exists x)(x + y = 20)$

3. Show if the following argument is valid or not

$$\begin{array}{l} p \\ \neg p \wedge q \\ \hline\hline q \end{array}$$

4. Let A, B, and C be subsets of a universal set U.

   (a) Prove the following set identity using the element method (i.e., show that each side is a subset of the other):

$$A \cap (B \setminus C) = (A \cap B) \setminus C$$

   (b) For the following statement, either prove it is true for all sets A, B, and C, or provide a specific counterexample to show it is false.

      i. $A \setminus (B \cap C) = (A \setminus B) \cap (A \setminus C)$

   (c) The *symmetric difference* of two sets A and B is defined as $A \triangle B = (A \setminus B) \cup (B \setminus A)$. Prove that $(A \triangle B) \triangle C = A \triangle (B \triangle C)$. (You may use a Venn diagram for insight, but your final proof must be analytical.)

5. (a) Use the Principle of Mathematical Induction to prove that for every natural number $n \geq 1$,

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}.$$

   (b) Let S be a non-empty set of real numbers that is bounded above. Define the lub of S. Let $T = \{x \in \mathbb{R} : -x \in S\}$. Prove that $glb(T) = -lub(S)$.

6. (a) Solve the equation $z^3 = -8i$. Express your answers in polar form ($re^{i\theta}$) and then convert them to Cartesian form ($a + bi$). Sketch the solutions on the complex plane.

   (b) Let $z = 2 + 2i$ and $w = 1 - \sqrt{3}i$.

      i. Find $\frac{z}{w}$ and express the result in exponential form $re^{i\theta}$.

      ii. Use De Moivre's Theorem to compute w^6, showing your work.

   (c) Describe geometrically the set of all complex numbers z that satisfy the equation:

$$|z - 3i| = 2|z + 1|$$